

SEMESTER END EXAMINATIONS – FEB / MARCH 2025

Program	: B.E. – Information Science and Engineering	Semester	: V
Course Name	: Computer Networks	Max. Marks	: 100
Course Code	: IS53 (2022 Batch)	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Define Data Communication and discuss its key characteristics. With a neat diagram illustrate the five components of data communication. CO1 (08)
 - For each of the following four networks, discuss the consequences if a connection fails. CO1 (04)
 - Five devices arranged in a mesh topology
 - Five devices arranged in a star topology (not counting the hub)
 - Five devices arranged in a bus topology
 - Five devices arranged in a ring topology
 - For the given 8-bit data 10110010, draw the line coding scheme for the following. CO1 (08)
 - Polar NRZ-L Scheme.
 - Polar RZ Scheme.
 - Manchester Scheme.
 - Pseudoternary Scheme.
- Describe the mesh and ring topologies in networking. Illustrate the structures and connections of each topology with a neat diagram. CO1 (08)
 - Define transmission impairment and discuss its causes with necessary diagrams. CO1 (09)
 - Define Nyquist Bit Rate. Calculate maximum for a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. CO1 (03)

UNIT - II

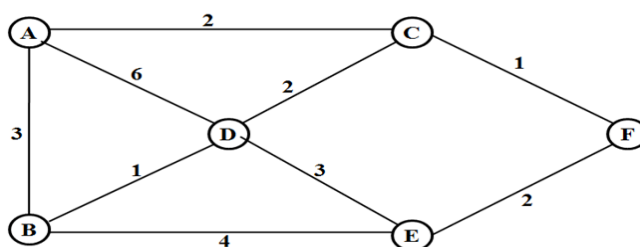
- Explain the Address Resolution Protocol (ARP). Discuss the process of ARP in resolving IP addresses to MAC addresses, and provide details about the structure of ARP packets, including the fields they contain. CO2 (10)
 - Given a dataword 1001 and divisor 1011, calculate the syndrome using the Cyclic Redundancy Check (CRC) method. Determine whether any errors are present in the transmitted data. CO2 (10)
- List and explain various types of Media Access Control(MAC) protocols. CO2 (10)
 - Find the codeword by applying CRC polynomial method. Given dataword as x^6+x^3 and divisor as x^3+x+1 CO2 (05)
 - Compute the checksum for the following data words at the sender's side: 0xDAAB, 0x7C1F, 0x67CE, and 0x9C99. CO2 (05)

UNIT - III

5. a) An ISP is granted a block of addresses starting with 120.50.0.0/16. The ISP wants to distribute these blocks to 2500 customers as follows:
- The first group has 2000 medium-size businesses; each needs 16 addresses.
 - The second group has 500 small businesses; each needs 8 addresses.

Design the subblocks and give the slash notation for each subblocks (*Mention the address range of First and Last customer in each group*). Find out how many addresses are still available after this allocation.

- b) Apply Bellman Ford routing to calculate the shortest path to the destination node B from all other nodes in the following network configuration.



- c) Describe the different open loop congestion control techniques. CO3 (04)
- d) A packet has arrived with an M bit value of 1. Is this the first fragment, the last fragment or a middle fragment? Comment on the additional flag bits required to answer the same. CO3 (02)
6. a) Briefly describe with necessary diagrams, the different decision-making strategies used by multicast distance vector routing. CO3 (08)
- b) What are the restrictions imposed on classless address blocks? An Organization is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets.
- Find the subnet mask
 - Find the number of addresses in each subnet
 - Find the network address of subnet 3.
- c) With a neat header diagram explain the different fields of an IPV4 datagram. CO3 (06)

UNIT- IV

7. a) Demonstrate with suitable flow diagrams, how TCP handles lost segments. CO4 (10)
- b) Discuss the different techniques used in the transition from IPV4 to IPV6 CO4 (10)
8. a) With a neat FSM diagrams, demonstrate the working of Go-Back-N protocol. CO4 (10)
- b) Compare IPv4 and IPV6 protocols with respect to their header formats. CO4 (07)
- c) Differentiate process of fragmentation with respect to IPV4 and IPV6 protocol. CO4 (03)

UNIT - V

9. a) Explain the TCP segment structure with a neat diagram. CO5 (10)
- b) Analyse how TCP establishes and terminates connection using three way hand shaking protocol. CO5 (10)
10. a) Briefly explain the two techniques used by DNS in address resolution. CO5 (10)
- b) Discuss how TCP handles congestion using Slow start, Congestion avoidance and Fast recovery methods. CO5 (10)
